

RIYADH KING KHALED INTL, Saudi Arabia

WMO: 404370

Lat: 24.958N

Lon: 46.699E

Elev: 2049

StdP: 13.64

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.6	43.2	6.7	8.4	63.3	10.8	10.3	67.0	20.5	62.3	18.6	63.1	3.4	300	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.5	112.8	66.0	111.2	65.5	109.7	65.2	69.9	100.9	68.3	99.8	67.1	100.3	9.5	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	92.8	71.8	61.0	86.4	70.6	58.9	80.2	70.5	35.1	100.3	33.7	99.6	32.7	99.3	76.8

Extreme Annual Design Conditions

Extreme Annual WS							Extreme Annual Temperature										n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years										
Min	Max				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max										
(a)	(b)	(c)			(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)										
19.9	17.6	15.2			34.2	115.3	3.7	1.4	31.6	116.3	29.4	117.1	27.4	117.9	24.7	118.9										
			DB		28.4	73.3	3.6	2.2	25.8	74.9	23.7	76.2	21.7	77.4	19.1	79.0										
			WB																							

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.1	58.0	62.7	70.0	79.5	89.6	94.3	96.4	96.1	90.5	81.4	68.9	60.3
	DBStd	14.75	5.96	6.58	6.23	5.96	4.23	2.78	2.66	2.54	3.52	4.10	5.95	5.90
	HDD50	13	8	2	0	0	0	0	0	0	0	0	0	3
	HDD65	559	227	112	23	1	0	0	0	0	0	0	29	167
	CDD50	10625	257	357	621	886	1229	1330	1438	1430	1214	973	567	323
	CDD65	5697	11	47	179	438	764	880	973	965	764	508	146	22
	CDH74	91638	154	498	1886	5469	12270	15527	17556	17275	12535	6771	1471	226
	CDH80	62369	19	130	744	2956	8220	11282	13108	12836	8598	3923	527	26
Wind	WSAvg	6.4	5.9	6.8	7.2	7.5	7.0	7.6	7.8	6.7	5.2	4.5	5.3	5.4
Precipitation	PrecAvg	4.6	0.7	0.3	0.8	1.0	0.3	0.0	0.0	0.0	0.0	0.0	0.7	0.7
	PrecMax	11.1	3.0	1.6	3.8	3.3	1.9	0.0	0.0	0.1	0.0	0.9	6.0	4.3
	PrecMin	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.6	0.8	0.4	1.1	0.7	0.5	0.0	0.0	0.0	0.0	0.2	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.4	88.1	95.3	102.9	109.9	113.3	114.9	114.1	110.9	102.5	91.8	82.8
		MCWB	57.3	59.3	59.8	62.6	64.5	65.9	66.7	67.4	66.1	63.6	60.8	58.1
	2%	DB	78.5	84.2	91.4	100.1	107.6	111.3	112.9	112.4	108.0	100.1	89.2	79.5
		MCWB	55.2	57.7	59.4	62.2	64.5	65.1	65.9	66.3	65.2	62.3	59.9	57.2
	5%	DB	74.8	80.3	87.8	96.8	105.5	109.5	111.3	111.0	106.1	98.0	86.1	76.5
		MCWB	53.8	56.1	58.5	61.7	64.2	64.5	65.3	65.9	64.5	61.7	59.2	56.3
	10%	DB	70.6	76.6	84.1	93.5	103.2	107.7	109.6	109.4	104.3	95.3	82.7	73.0
		MCWB	52.8	54.4	57.4	61.0	63.8	63.9	65.0	65.7	64.0	61.1	58.2	55.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.9	63.5	66.4	69.1	69.9	69.3	71.7	72.8	70.0	68.7	67.4	64.3
		MCDB	70.8	77.1	79.9	83.3	95.1	106.1	104.3	102.1	101.1	86.8	76.6	72.5
	2%	WB	59.2	60.8	63.8	66.8	67.9	67.0	68.5	69.9	67.5	66.2	64.6	61.7
		MCDB	69.0	74.7	79.2	83.0	95.4	106.7	107.5	104.7	101.3	87.0	75.9	70.9
	5%	WB	57.0	58.7	61.9	65.2	66.4	65.6	66.9	68.1	66.1	64.5	62.9	59.6
		MCDB	68.6	73.7	78.0	83.5	95.9	106.7	107.7	106.0	101.9	88.7	75.8	69.5
	10%	WB	54.8	56.5	59.9	63.8	65.1	64.5	65.6	66.7	64.9	62.8	61.3	57.5
		MCDB	67.6	72.9	77.1	83.8	96.3	106.0	107.0	106.1	101.4	90.4	75.4	68.9
Mean Daily Temperature Range	5% DB	MDBR	21.9	22.9	23.5	24.2	25.5	27.7	27.5	27.7	28.2	27.4	22.6	21.9
		MCDBR	27.3	27.1	28.0	27.7	27.9	29.2	28.3	28.7	29.0	29.2	27.6	26.4
		MCWBR	13.1	12.4	11.7	9.8	9.8	10.5	10.2	10.2	10.7	11.3	11.7	11.7
	5% WB	MCDBR	21.2	22.6	22.9	23.2	24.9	27.7	27.4	27.1	27.4	24.8	19.7	18.7
		MCWBR	11.4	11.6	10.8	8.9	9.0	10.3	10.6	10.5	10.9	10.9	8.9	9.3
Clear-Sky Solar Irradiance	taub	0.429	0.474	0.543	0.627	0.649	0.605	0.584	0.575	0.519	0.455	0.467	0.427	
	taud	1.974	1.846	1.690	1.563	1.522	1.580	1.648	1.666	1.761	1.908	1.888	1.976	
	Ebn at Noon	256	253	243	226	221	229	234	235	246	255	240	250	
	Edh at Noon	51	62	76	89	92	87	81	79	70	57	55	49	
All-Sky Solar Radiation	RadAvg	1355	1627	1871	2034	2273	2489	2381	2263	2078	1801	1385	1282	
	RadStd	95	91	106	103	90	79	54	50	49	70	102	86	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (1 neighbor)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	+202	+185	

Nomenclature: See separate page